# Gazipur Digital University

*Department of Software Engineering*

# SDLC Selection

*for*

***"ReVendo: Buy, Sell & Repair Pre-Owned Items"***

**Submitted by:**

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# SDLC Model Selection for ReVendo Project

## Project Context Analysis

**Project**: ReVendo - Buy, Sell & Repair Pre-Owned Items
**Team**: Trivendo (3 members)
**Timeline**: 9 weeks(est.)
**Current Status**: Console application prototype planned.

## Selection Criteria and Priorities

Based on ReVendo's project characteristics, the following criteria are most important:

| Criteria | Priority Weight | Rationale |
|---|---|---|
| Requirements Clarity | High (9) | SRS is comprehensive and well-defined |
| Team Size & Experience | High (8) | Small team (3 members), student project |
| Timeline Constraints | High (9) | Fixed 9-week academic schedule |
| Quality Assurance | High (8) | Marketplace requires reliability and security |
| User Feedback Integration | Medium (7) | Need stakeholder validation for UI/UX |
| Risk Management | Medium (6) | Moderate technical and operational risks |
| Flexibility for Changes | Medium (5) | Requirements mostly stable. |

## SDLC Model Comparison Matrix

| Criteria | Weight | Waterfall | V-Model | Iterative | Spiral | Agile | Prototyping |
|---|---|---|---|---|---|---|---|
| Requirements Clarity | 9 | 9 | 9 | 7 | 6 | 5 | 4 |
| Team Size & Experience | 8 | 8 | 8 | 6 | 4 | 6 | 7 |
| Timeline Constraints | 9 | 8 | 7 | 6 | 4 | 7 | 5 |
| Quality Assurance | 8 | 6 | 9 | 7 | 8 | 6 | 5 |
| User Feedback Integration | 7 | 3 | 4 | 7 | 8 | 9 | 9 |
| Risk Management | 6 | 4 | 6 | 7 | 9 | 7 | 6 |
| Flexibility for Changes | 5 | 2 | 3 | 8 | 9 | 9 | 8 |

# Weighted Scores Calculation

| Model | Calculation | Total Score |
|---|---|---|
| Waterfall | (9×9)+(8×8)+(9×8)+(8×6)+(7×3)+(6×4)+(5×2) | **249** |
| V-Model | (9×9)+(8×8)+(9×7)+(8×9)+(7×4)+(6×6)+(5×3) | **268** |
| Iterative | (9×7)+(8×6)+(9×6)+(8×7)+(7×7)+(6×7)+(5×8) | **248** |
| Spiral | (9×6)+(8×4)+(9×4)+(8×8)+(7×8)+(6×9)+(5×9) | **243** |
| Agile | (9×5)+(8×6)+(9×7)+(8×6)+(7×9)+(6×7)+(5×9) | **252** |
| Prototyping | (9×4)+(8×7)+(9×5)+(8×5)+(7×9)+(6×6)+(5×8) | **238** |

# Recommendation: V-Model

**Selected Model**: **V-Model (Verification & Validation Model)**
**Score**: 268/315 (85%)

The V-Model is the precise choice for ReVendo because:

**Strengths Matching Project Needs:**

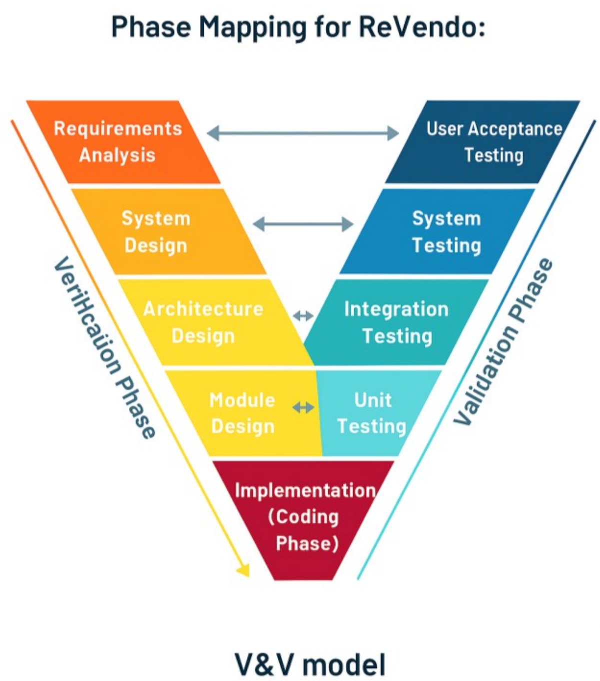
1. **Clear Requirements Match**: ReVendo has well-documented SRS with stable and understood requirements.
2. **Quality Focus**: Marketplace applications require high reliability any day. V Model's parallel testing approach ensures quality production.
3. **Team Suitability**: Perfect for small and organized teams with clear role definitions and we're a 3 members team in Trivendo.
4. **Academic Timeline**: Structured phases align perfectly with our estimated 9-week academic schedule.
5. **Verification Emphasis**: Each development phase has corresponding testing phase.

**Specific Benefits for ReVendo:**

- **Security Testing**: Parallel testing approach ascertains user data protection and transaction security which actually is one of the main mottos of the project.
- **Integration Testing**: Critical for marketplace components (user management, product catalog, messaging, eco-tracking).
- **Documentation**: Strong documentation practices satisfy future maintenance.
- **Milestone Clarity**: Clear phase boundaries help track progress within a fixed timeline.

*Please Turn Over*

### Implementation Strategy




### Why Other Models Were Not Selected?

Among all the SDLC approaches we've calculated, Agile scored well but the fixed deliverables and timelines of the academic environment favor more structured methods. Waterfall has insufficient quality assurance for marketplace security, while Iterative offers flexibility but less emphasis on testing compared to the V-Model. Spiral is unnecessarily complex for this project's scope and timeline, and Prototyping is unnecessary since requirements are already ascertained.

# Conclusion

The V-Model offers a good balance of structure, quality checks, and schedule management for the ReVendo project, making it the best SDLC approach for delivering the targeted circular economy marketplace.